

Symphonix-Health CRAFTS agentic intelligence architecture

Revised against the local Symphonix repositories. The prior draft treated several repos as inaccessible; this edition replaces that constraint with code and specification evidence.



Working interpretation: Symphonix-Health is best described as governed, networked clinical intelligence infrastructure. It is not evidenced as a generic AGI platform. "AGI" can remain a research horizon term, but the architecture that exists is a bounded clinical agent network with identity, consent, routing, safety, and evidence gates.

CRAFTS status: CRAFTS is used here as an analytical lens: Culture, Relations, Agency, Funding or resource governance, Technology, and Structure. It is not a runtime subsystem in the codebase.

CORRECTION

Repo access is no longer treated as a blocker. GHARRA, Nexus A2A, BulletTrain, Bridge SDK, Prompt Engine, CSAA, SignalBox, CAID, and Patient360 provide direct evidence.

REMAINING GAP

The missing object is not a persona registry. It is a single cross-plane projection binding persona, skills, tools, IAM, safety, budgets, consent, and evaluations.

- BulletTrain data plane
- GHARRA trust registry
- Nexus A2A
- Bridge SDK contracts
- CSAA safety cases
- SignalBox attestation

Primary revisions from the incomplete source report

The source PDF's strongest limitation was its claim that local repositories were unavailable. This revision treats the public narrative as secondary and uses the local repos as the control surface.

Source draft position	Repo-aligned correction	Evidence status
Architecture is "spec-verified and interface-verified, not fully code-verified."	Core claims are now code/spec verified across BulletTrain, GHARRA, Nexus, Bridge SDK, Prompt Engine, CSAA, SignalBox, CAID, and Patient360.	closed Local repositories are readable.
Unified persona registry may be missing.	GHARRA has a `PersonaRegistry` and a `PersonaDefinition` surface that consolidates 68 persona definitions and country-aware avatar selection.	reframed Cross-repo contract projection is still partial.
Specialty persona completeness is not verified.	Local persona code includes psychiatrist, dietitian, pharmacist, and nurse practitioner examples with toolsets and FHIR role codes.	verified Completeness of every specialty remains a testable inventory task.
Compute power should be added as a new CRAFTS dimension.	The code supports a narrower term: resource governance. Bridge SDK defines per-persona and per-patient budgets; Bevan and Prompt Engine route model work through governed gateways.	refined No evidence of an infrastructure ownership layer.
Computational intuition can be treated as a realized layer.	The repos show planning, reflection, routing, provider fallback, and capability discovery, but no single global "intuition" service.	design pattern Needs explicit contract and tests if adopted.

Wording change
Use "networked clinical intelligence" for the current architecture. Reserve "networked superhuman clinical intelligence" for a carefully bounded thesis claim, not for code readiness.

Diagram change
The revised diagrams put BulletTrain, Bridge SDK, GHARRA, Nexus, Prompt Engine, Bevan, CSAA, SignalBox, and CAID into explicit layers instead of showing a generic agentic stack.

Repo-backed architecture layers

The codebase supports a layered clinical intelligence architecture. Data, identity, reasoning, execution, and evidence are separate concerns.

External substrate	Health systems, national rails, payers, pharmacies, labs, imaging, ambulance workflows, and external model providers are dependencies, not proof of autonomy.	<code>Kenya SRS</code> <code>Bridge adapters</code> <code>BulletTrain</code>
Data plane	Cross-system data movement belongs in BulletTrain. Bridge SDK translates external protocols and projects consent, provenance, FHIR tasks, and audit records.	<code>BulletTrain</code> <code>symphonix-bridge-sdk</code>
Model and reasoning plane	Prompt Engine and Bevan route model calls through governed gateways with policy prompts, schema validation, PHI boundaries, fallback, and output checks.	<code>prompt-engine</code> <code>BulletTrain/bevan_llm</code>
Trust and route plane	GHARRA provides zero-trust discovery and trust metadata. Nexus validates agent records and runs A2A route admission and task lifecycle.	<code>global-agent-registry</code> <code>nexus-a2a-protocol</code>
Persona and contract plane	Personas are defined in GHARRA and surfaced through Nexus, Bridge SDK, Patient360, and BulletTrain governance personas. This is the right home for superpersona contracts.	<code>personas.py</code> <code>agent_contract.py</code> <code>Patient360</code>
Governance and evidence plane	CSAA, SignalBox, CAID, RBAC, audit stores, HITL coordination, and real-service test rules decide whether an agent path can move beyond shadow or assisted operation.	<code>csaa</code> <code>signalbox-mcp</code> <code>caid-agent</code>

CRAFTS after code verification

CRAFTS remains useful, but each dimension should map to real components and contracts. The categories below avoid treating social theory terms as deployed services.

Culture
Clinical norms, safety language, national constraints, consent posture, and model behavior rules.
Prompt Engine policy prompts; CSAA; Kenya SRS; design invariants.

Relations
How agents, humans, tenants, patients, and systems can see, trust, and route to one another.
GHARRA; Nexus route admission; Bridge consent/provenance; Patient360 context radius.

Agency
Bounded capacity to propose, draft, classify, route, summarize, or escalate clinical and operational work.
68 persona definitions; 25 Nexus agents; Bridge typed intents; HITL workflows.

Funding and resource governance
Resource ceilings, model routing, concurrency, queues, and cost-sensitive routing, not a broad funding theory.
Bridge concurrency budgets; Bevan routing/fallback; workflow metrics.

Technology
The actual data, protocol, model, agent, and observability systems that make the network executable.
BulletTrain; Bridge SDK; Prompt Engine; Bevan; SignalBox; CAID.

Structure
Governance boundaries, role permissions, safety cases, audit duties, and release gates.
RBAC; GHARRA ABAC; CSAA hard stop; real-service evidence rules.

Mechanism claim: Symphonix changes the clinical decision environment by making data movement, identity, persona scope, model routing, safety, and evidence visible and governable. The mechanism is the governed network, not a free-floating agent.

Dimension	Mechanism class	Boundary condition
Culture	Norms and policy prompts shape acceptable clinical behavior.	Policy text is not enough without tests and safety cases.
Relations	Trust metadata and consent projection decide which exchanges can happen.	GHARRA remains no-PHI; PHI routing must stay local.
Agency	Persona-scoped intents and tools make agent action bounded.	Legal or clinical effect needs human approval unless evidence says otherwise.

The persona gap moved

The old report treated a unified persona registry as likely missing. Local code changes that conclusion: GHARRA has a consolidated persona registry. The live gap is cross-plane projection from persona to skills, tools, IAM, safety, budgets, consent, and evaluations.

Verified persona surface

- GHARRA `PersonaDefinition` includes key, label, category, summary, toolset, SOPs, domains, FHIR role code, and avatar variants.
- `PersonaRegistry` exposes the consolidated persona catalog.
- The code comments state that the module consolidates three persona systems, 68 persona definitions, country-aware avatar selection, and capability scoping.

Specialty examples now verified

- Psychiatrist persona exists with clinical toolset and FHIR role code.
- Dietitian persona exists with nutrition toolset and FHIR role code.
- Pharmacy personas and nurse practitioner personas are present.

Plane	Current evidence	Cross-plane risk
GHARRA	Persona definitions, avatar variants, capability scoping, FHIR role codes, no-PHI trust boundary.	Registry does not by itself prove runtime authorization.
Nexus	25 clinical agents, 68 personas, six IAM groups, A2A route admission, JSON-RPC task lifecycle.	Needs generated contract alignment with GHARRA personas.
Bridge SDK	Typed agent intents, safety class, FHIR Task projection, persona delegation token fields, agent-card output.	Needs persona catalog source-of-truth linkage.
Patient360 and BulletTrain	Persona-centered patient context, FCG personas, HITL orchestration, clinical workflow governance.	Needs evidence that UI roles, data radius, and backend contracts agree.

A canonical superpersona contract should be generated, not hand-described

The architecture has most of the necessary pieces. The next refactor target is a generated contract that projects a persona into every plane that can authorize, route, execute, observe, or test it.

persona_id
GHARRA PersonaRegistry

human_role
FHIR role code; IAM group

skill_pack
persona toolset; SOPs

tool_pack
Bridge SDK capabilities

safety_class
Bridge AgentContract; CSAA

jurisdiction_pack
Kenya SRS; GHARRA variants

consent_projection
Bridge
consent_projection.py

budget_limits
Bridge concurrency.py

route_admission
Nexus A2A; GHARRA metadata

model_policy
Prompt Engine; Bevan
router

eval_pack
SignalBox; CAID; scenario
YAML

audit_pack
Bridge audit; BulletTrain
audit store

Contract field	Exists locally?	Needed revision
Persona identity and clinical role	yes GHARRA persona definitions and Nexus IAM groups.	Generate one canonical ID and alias table consumed by every sibling repo.
Skill and tool authorization	partial Persona toolsets and Bridge capabilities exist.	Compile allowed tool calls per persona, task type, tenant, and jurisdiction.
Safety and evaluation binding	partial Safety classes, CSAA, SignalBox, and CAID exist.	Make deployment fail when the persona lacks a passing safety case and scenario pack.
Budget and compute limits	partial Bridge concurrency budgets exist.	Expose budgets as a first-class agent contract field and report saturation telemetry.

"Compute power" is better modeled as resource governance

The source report proposed compute power as a possible extra CRAFTS dimension. The repos support a narrower and more useful architecture term: resource governance.

What is code-backed

- Bridge SDK has per-persona and per-patient execution budgets with saturation behavior and limit dimensions.
- Bevan and Prompt Engine route model calls through governed gateways rather than letting every sibling call arbitrary models directly.
- GHARRA documentation preserves a no-PHI boundary; PHI model routing belongs inside local Bevan-controlled paths.
- HITL queue metrics and workflow gates provide operational back-pressure.

What is not evidenced

- No local evidence was found for an estate-level GPU ownership or data-center control layer.
- No single service currently declares all persona budgets, model budgets, queue limits, and safety gates as one contract.
- Therefore the diagram should not imply compute sovereignty beyond the code-backed routing and budget controls.

1

Task arrives

Persona, tenant, patient, task, consent, and jurisdiction context are identified.

2

Budget check

Per-persona and per-patient limits decide whether the task can execute now.

3

Route choice

Prompt Engine and Bevan choose local or external model paths under policy.

4

Safety gate

CSAA, HITL policy, output validation, and consent rules constrain action.

5

Evidence

Audit, SignalBox scenarios, CAID checks, and metrics decide readiness.

Revised CRAFTS wording: keep Funding as a social and procurement context, but add "resource governance" as the deployed system dimension that CRAFTS must inspect for agentic healthcare.

Computational intuition is a proposed pattern, not a verified global service

The idea is still useful if it is defined as fast pre-planning: salience detection, route choice, likely framework selection, risk spotting, and tool-pack selection before deeper execution. The current repos show pieces of that pattern, not a single deployed intuition layer.

Candidate function	Repo-backed evidence	Required to claim it as a layer
Fast salience detection	Prompt Engine detects clinical/PHI patterns and can auto-activate HIPAA, safety, bias, and citation prompt elements.	Explicit salience schema and tests across clinical pathways.
Framework selection	Persona definitions, SOPs, toolsets, and clinical safety classes provide the vocabulary.	A deterministic selector that records why a framework was chosen.
Route and model choice	Bevan and Prompt Engine support governed LLM routing, fallback, and output checks.	A route decision trace bound to persona, consent, and safety case.
Risk spotting	CSAA can hard-stop unresolved residual risks; Bridge SDK has safety classes and clinical safety cases.	Risk classifier evidence tied to SignalBox/CAID scenarios.

Recommended diagram label

"Policy and routing heuristics" is more accurate than "computational intuition" for the current architecture.

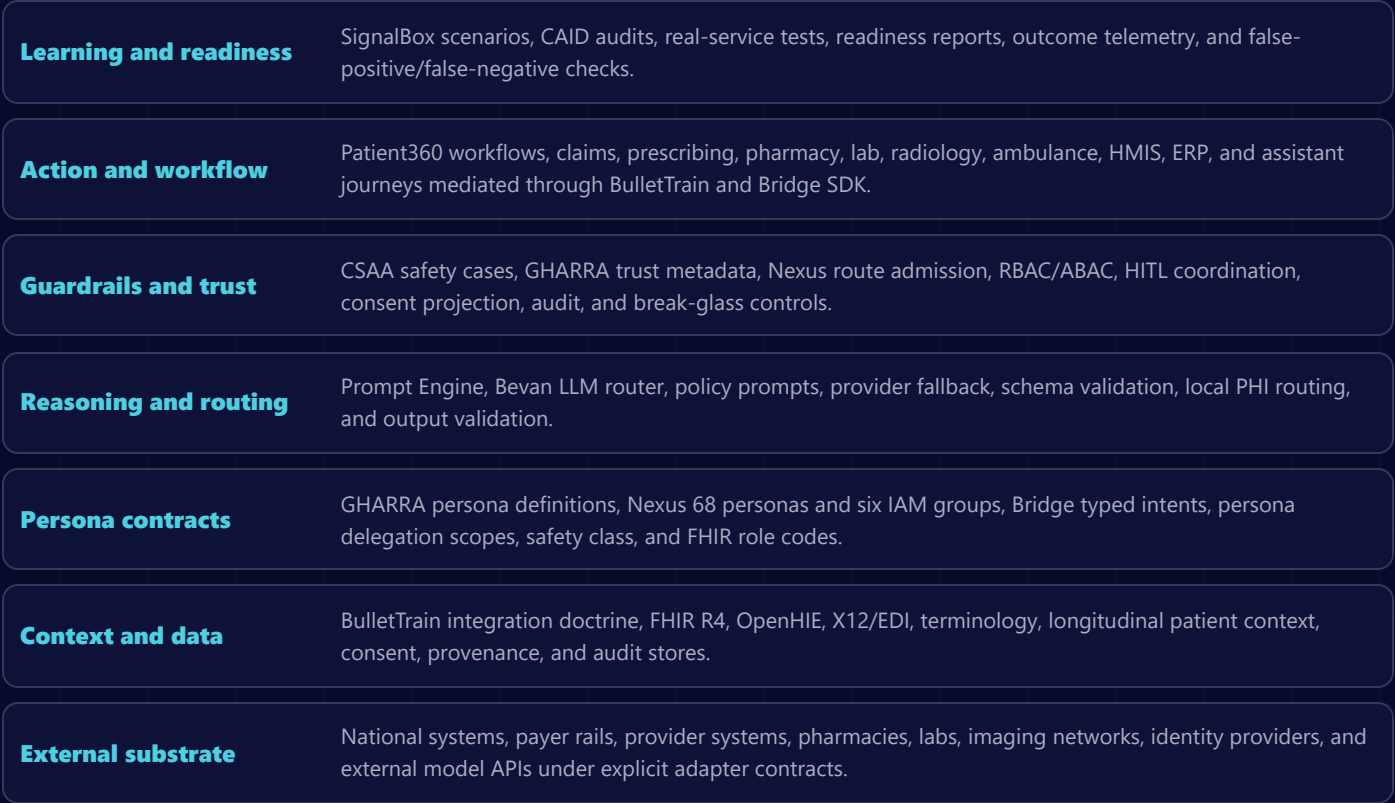
Recommended implementation

If this becomes a product layer, implement it as a typed decision record: input context, selected persona, selected tools, model path, risk class, and evidence pack.

Do not imply autonomous clinical diagnosis from this layer. The verified behavior is bounded recommendation, drafting, routing, classification, summarization, and escalation under human and safety governance.

Corrected nested architecture

The nested stack below replaces a generic agentic diagram with repo-backed layers. Each layer depends on the layers below it and is constrained by safety and evidence above it.



The diagram should be read bottom-up for dependency and top-down for governance. No agent bypasses the data, consent, trust, safety, and evidence layers.

Roadmap after repo access was restored

The roadmap no longer starts with "restore repository access." It starts with turning already-present surfaces into generated contracts and executable gates.

Workstream	Status	Next action
Persona inventory	partly closed	Generate a normalized persona inventory from GHARRA, Nexus, Patient360, Bridge SDK, and BulletTrain role files. Flag aliases and missing mappings.
Superpersona contract	partial	Generate a `superpersona.contract.json` per persona with skill, tool, IAM, safety, consent, budget, route, model, and evaluation fields.
Safety as CI gate	partial	Make CSAA unresolved risk, missing SignalBox scenario, missing CAID check, or missing real-service journey fail readiness.
Resource governance	partial	Project Bridge concurrency budgets and Bevan routing limits into persona contracts and telemetry dashboards.
Architecture catalog	gap	Create a cross-repo generated map of services, routes, persona roles, data flows, PHI boundaries, and evidence artifacts.
Outcome learning loop	gap	Bind outcome metrics to approved workflow changes, prompt updates, safety case revisions, and persona contract versions.

1

Inventory
Extract persona, route, contract, safety, and scenario surfaces from each repo.

2

Normalize
Resolve aliases, role codes, IAM groups, tenant scope, and jurisdiction variants.

3

Generate
Emit superpersona contracts and architecture catalog artifacts.

4

Gate
Require safety, real-service, and scenario evidence before readiness claims.

5

Learn
Feed verified outcomes into controlled contract and policy updates.

Rules for future diagrams and reports

Future architecture documents should use these checks before claiming readiness, coverage, or agentic maturity.

Evidence rules

- Use local repo code/specs as the control source before public pages or narrative reports.
- Distinguish code evidence, requirement evidence, scenario evidence, and inference.
- Do not count reports, dashboards, or generated matrices as verdicts without source checks.
- Run false-positive and false-negative checks for disputed coverage claims.

Clinical readiness rules

- Use real seeded service paths for internal systems; do not replace reachable internal services with mocks as readiness evidence.
- No agent path should bypass consent, provenance, audit, PHI boundaries, HITL policy, route admission, and safety cases.
- For backend contracts, UI evidence alone is incomplete unless network/backend evidence is also captured.
- Break-glass, emergency, and legal-effect workflows need explicit human control and audit evidence.

Claim type	Minimum evidence	Bad shortcut to avoid
Persona exists	Registry entry, role code, alias mapping, IAM mapping, and runtime route or UI use.	Counting a name in one markdown file.
Agent can act	Typed intent, allowed tools, consent, safety class, route admission, budget, audit, and tests.	Inferring action from a model prompt alone.
Workflow is ready	Real seeded data path, backend and UI checks, SignalBox/CAID evidence, and safety case status.	Scenario-only pass without service evidence.
Architecture is integrated	Cross-repo contract map, route map, service ownership map, and data/provenance flow.	One diagram with no generated ledger.

Repo evidence used for this revision

Repo/file	Lines	Evidence used
symphonix-health-docs/strategy/agent-first.md	9; 19-21; 33-42; 54-59; 193-209	Agent-first sibling roles, API boundaries, CSAA/GHARRA/Nexus control model, discoverable/reachable/grounded/gated/observable/reversible requirements, HITL patterns.
symphonix-health-docs/strategy/bullettrain-integration-doctrine.md	11-23; 61-100; 117-119	BulletTrain as the required cross-system data movement plane; AI agents may recommend/draft/classify/route but cannot mutate sibling systems outside authorized auditable mediation.
BulletTrain/README.md	33; 43-69; 121-130; 183-219; 318-380	160+ microservices, FHIR/OpenHIE/X12/terminology support, SignalBox/FCG/governance, RBAC, HITL, Bevan, Patient360, context assembler, workflow orchestrator, adapters.
global-agent-registry/README.md	26; 93-119; 139; 165-166; 259-268	GHARRA zero-trust discovery/routing/trust registry, no-PHI boundary, ABAC, consent, FHIR R4, break-glass, agent cards, and dashboard.
global-agent-registry/src/gharra/core/personas.py	4-18; 187-199; 288-296; 524-533; 545-583; 625-633; 970-984	Consolidated persona registry, 68 persona definitions, country-aware avatar selection, capability scoping, persona fields, psychiatrist, nurse practitioner, pharmacy, dietitian, and `PersonaRegistry`.
nexus-a2a-protocol/README.md	26-30; 43; 137; 187-200; 222; 263	Trusted identity, AI Gateway, JSON-RPC, JWT, task lifecycle, 25 clinical agents, 68 personas, six IAM groups, identity endpoints, persona/IAM docs.
nexus-a2a-protocol/docs/gharra_integration.md	5; 14; 18-21; 228-231	Nexus consumes GHARRA trust/routing metadata, validates records, and does not replace GHARRA discovery or BulletTrain orchestration.
symphonix-bridge-sdk/src/bridge_sdk/agent_contract.py	1-13; 178-185; 233-248; 275-310	Agent Execution Contract, typed intents, safety class, FHIR Task conversion, GHARRA agent-card capability strings, and `.well-known/agent-card.json` output.
symphonix-bridge-sdk/src/bridge_sdk/concurrency.py	1-4; 23; 57-86; 102-182	Per-persona and per-patient execution budgets, saturation, dimensions, and limits.
symphonix-bridge-sdk/src/bridge_sdk/consent_projection.py	16-23; 95; 123-148	Consent projection travels on persona token and task, including jurisdiction and stale-consent checks.
prompt-engine/docs/REQUIREMENTS.md	344-363; 643-670; 950-956; 1026	Agentic mode, persistence/tool-use/planning, safety policy prompt elements, reflection/error recovery, PHI and clinical pattern detection.
prompt-engine/src/prompt_engine/optimizer/workflow.py	4-11; 227	Every LLM call routes through the BulletTrain API gateway hub and schema validation.
csaa; signalbox-mcp; caid-agent; patient360-assistant	CSAA 90-94, 149; SignalBox v2 217-229; CAID 245-249; Patient360 README 55-61, 128-132	Safety scope and hard-stop behavior, scenario evidence, FP/FN audit discipline, Patient360 clinical assistant and persona/context expectations.